

Domain, Why and Who

The visualisation examines the modern Olympic Games (1896–2024), focusing on global medal distribution and Australia's performance trajectory. It converts 128 years of Olympic data into an accessible narrative for the average Australian adult who follows sport casually and has no statistical background. Comparisons use only plain measures such as medal counts, nation rankings, and years without statistical jargon.

What?

The visualisation is based on two data sources. Source 1 is the Wikipedia All-Time Olympic Games Medal Table, compiled into *medals_data.csv*, containing cumulative gold, silver, and bronze medal counts by country. Source 2 is the IOC Official Historical Games Records, organised into *history_data.csv*, *host_data.csv*, *aus_perf.csv*, and *aus_sport.csv*, covering athlete participation, host cities, and Australia-specific performance data.

How?

1. **Choropleth Map** – Uses a log-scale sequential colour scheme on country geoshapes. The log scale prevents the USA's 2,938 medals from compressing variation among other nations into indistinguishable colours.
2. **Proportional Symbol Map** – Places scaled circles on host-city locations, encoding athlete count by area and Olympic season by colour, communicating geography, event scale, and temporal distribution simultaneously(custom-built element used here is buttons).
3. **Heatmap** – Uses a continent-by-decade matrix where colour saturation represents the number of Games hosted. Embedded text labels provide exact counts without requiring legend cross-referencing.
4. **Dumbbell Chart** – A custom three-layer design connecting gold and total medal points on a shared axis. The distance between points reveals medal conversion efficiency, a relationship not easily seen in a standard bar chart.
5. **Bubble Chart** – Encodes summer medals (x-axis), winter medals (y-axis), total medals (size), and continent (colour), with interactive continent filtering for multidimensional comparison(custom-built element used here is buttons).
6. **Stacked Bar Chart** – Uses colour to represent gold, silver, and bronze medals stacked on a common baseline, enabling part-to-whole comparisons among leading nations.
7. **Dual-Axis Growth Line Chart** – A custom three-layer chart with independent scales and suppressed duplicate axes. An area fill emphasises athlete-growth magnitude over time.
8. **Stacked Area Chart** – Displays gold, silver, and bronze medals as stacked areas across time, showing both total medal trends and medal composition.
9. **Dual-Axis Rank Chart** – Plots medal totals and national rank on separate y-axes, highlighting that increased medal counts do not always produce higher rankings when global competition intensifies.
10. **Diverging Bar Chart** – Uses dynamic calculations to compare each year against Australia's historical average. Bars extending above or below zero clearly indicate overperformance or underperformance without pre-processed data.

11. **Lollipop Chart** – Combines total medals and gold-medal contributions within a compact three-layer design, conveying both quantity and gold-medal ratio more efficiently than grouped bars.
12. **Donut Chart** – Uses arc segments to represent medals by sport category. The centre displays the overall total of 559 medals, turning unused space into a meaningful annotation that anchors proportional comparisons.